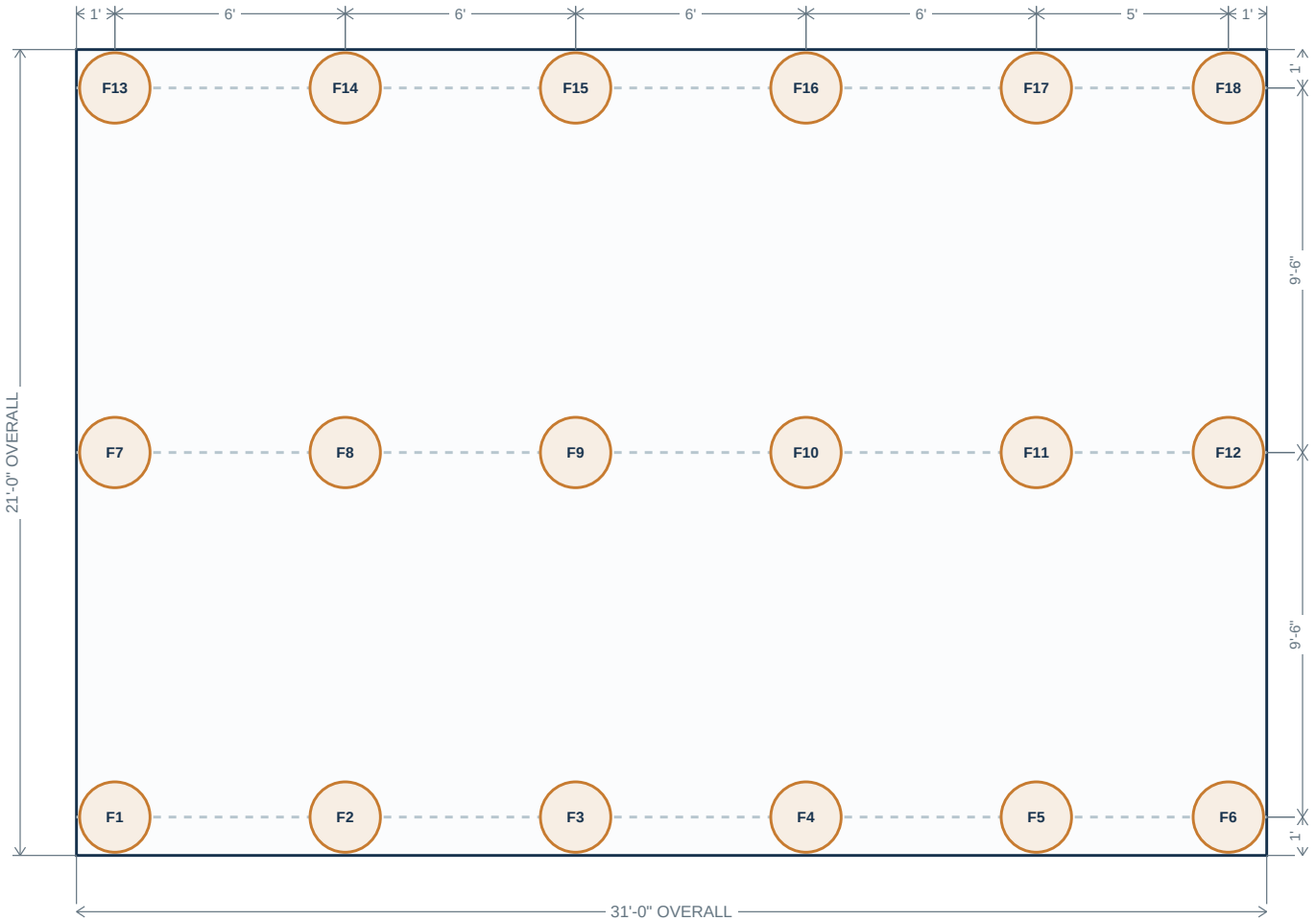
PIER AND FOOTING PLAN

Separate foundation layout with centerline coordinates and typical pier section

Blue Springs, Missouri | July 16, 2026

22-INCH PAD FOOTINGS - 18 TOTAL

Dashed lines = beam centerlines



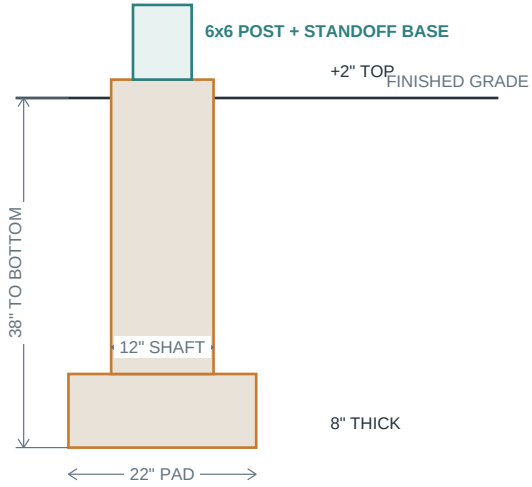
FOOTING CENTERLINES

Row	Footings	Y	X coordinates
South	F1-F6	1'-0"	1', 7', 13', 19', 25', 30'
Middle	F7-F12	10'-6"	1', 7', 13', 19', 25', 30'
North	F13-F18	20'-0"	1', 7', 13', 19', 25', 30'

FOUNDATION REQUIREMENTS

18 total piers; one under each post
22-in round x 8-in concrete pad
12-in-diameter shaft above pad
Pad bottom elevation: -38 in
Pier top elevation: +2 in
Assumed soil bearing: 1,500 psf

TYPICAL PIER SECTION



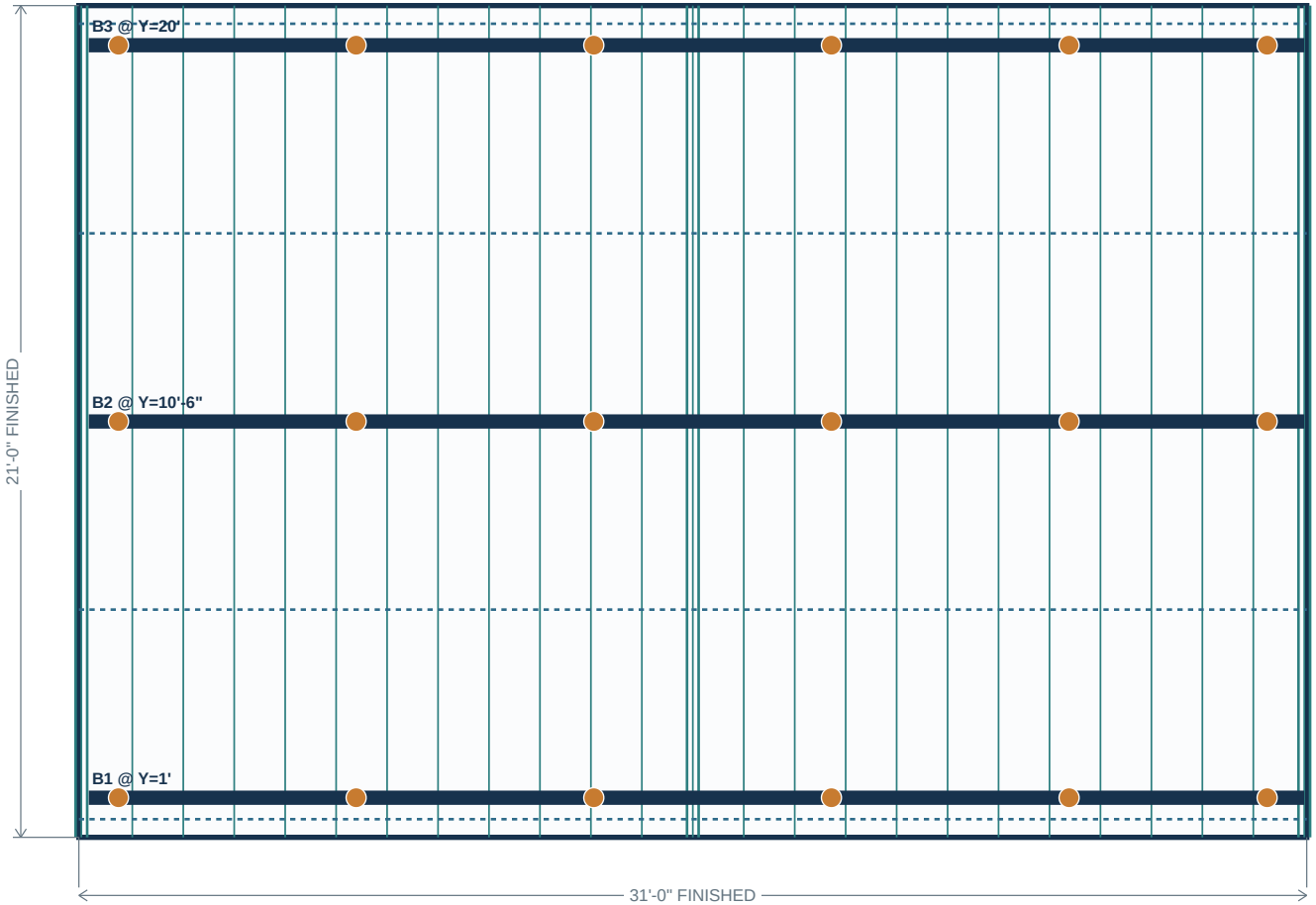
JOIST AND BEAM FRAMING PLAN

Flush-beam configuration, joist spacing, blocking, rims, posts, and connection concept

Blue Springs, Missouri | July 16, 2026

2x8 JOISTS RUN NORTH-SOUTH | 15-7/16" O.C.

Blue dashed = blocking | Orange = 6x6 post



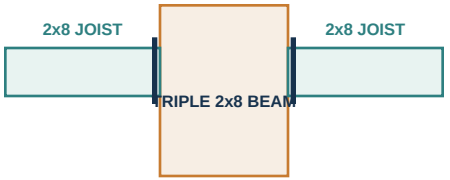
FRAMING SCHEDULE

Joists: PT SYP No. 2 2x8
Actual spacing: 15-7/16 in o.c.
Main clear spans: 9'-1-1/2 in
Beams: (3) 2x8 flush built-up
Posts: 18 PT 6x6, six per beam
Rims: PT 2x8 with 1x8 fascia
No joist cantilevers

BLOCKING AND BRACING

Solid row at each field midspan
Solid support below long borders
Bridge doubled joists <=24 in o.c.
8 corner knee braces, field fit
Two underside CS16 diagonal runs

FLUSH-BEAM CONNECTION CONCEPT



Joists terminate in listed LUS28Z hangers
Beam bears on 6x6 post through listed cap

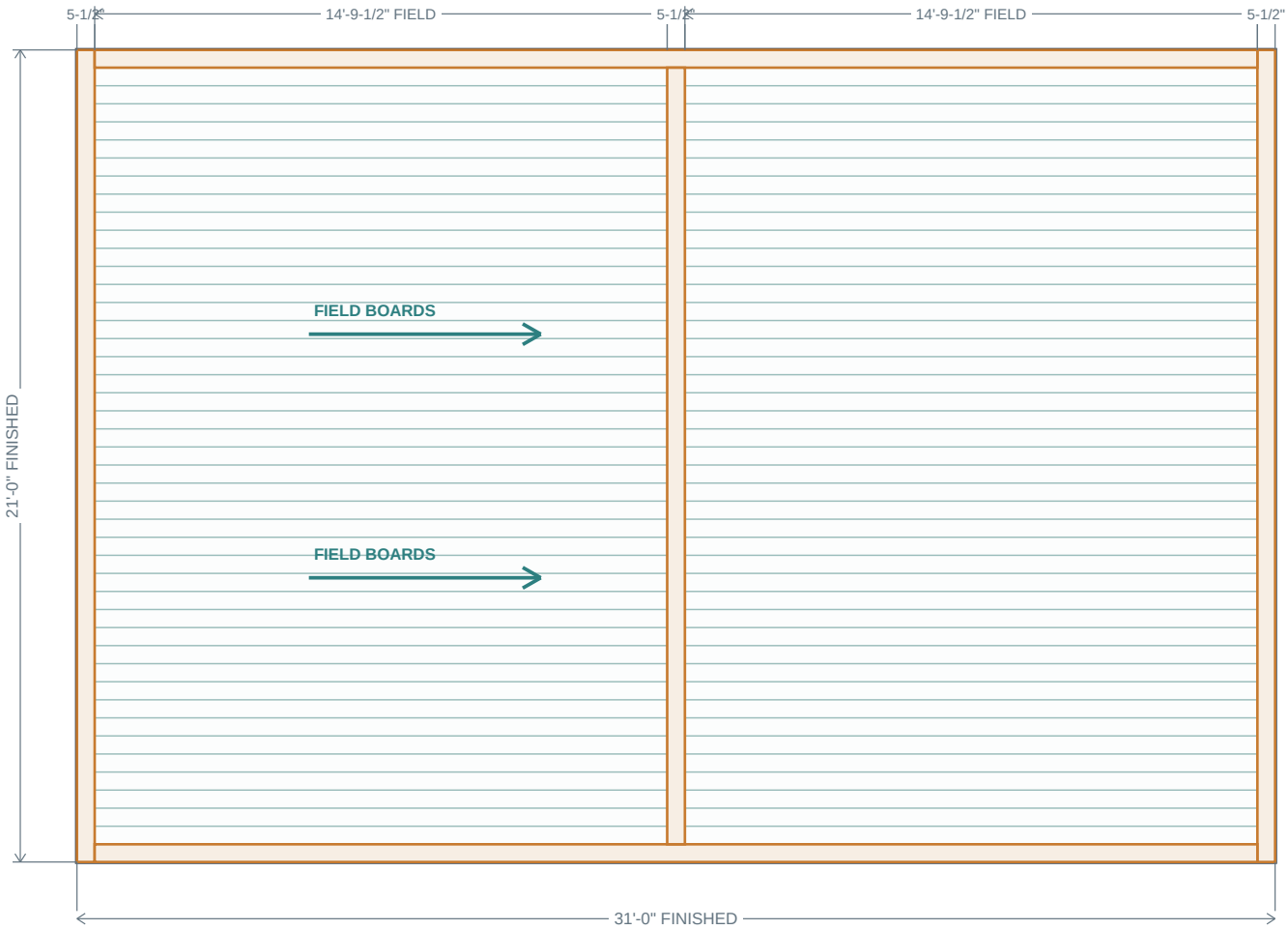
DECKING AND PICTURE-FRAME PLAN

Common board layout for Option A 5/4x6 or Option B 2x6 pressure-treated decking

Blue Springs, Missouri | July 16, 2026

ORANGE = 5-1/2-IN PICTURE FRAME / CENTER BREAKER

43 full field courses | boards run east-west



COMMON EXACT LAYOUT

Overall surface: exactly 31' x 21'
Three 5-1/2-in vertical control boards
Four 1/8-in longitudinal gaps
Each field piece: 14'-9-1/2 in
No random field butt joints
43 full field courses across width
Balance 44 gaps to total 4-1/2 in

OPTION A - 5/4x6

100 x 16-ft, 4 x 12-ft, 4 x 10-ft
Two #9 x 2-1/2-in screws/support
Post cut: 12-3/4 in
Recommended best-value option

OPTION B - 2x6

100 x 16-ft, 4 x 12-ft, 4 x 10-ft
Two #9/#10 x 3-in screws/support
Post cut: 12-1/4 in
Same footing and framing layout

Verify actual average board width before framing.
Use a story pole; do not rip a narrow final course.

MATERIALS LIST - COMMON STRUCTURE AND SITE

Purchase quantities include practical overage and apply to both decking options

Blue Springs, Missouri | July 16, 2026

COMMON PURCHASE SCHEDULE

Material	Purchase quantity	Primary use / calculation
PT SYP No. 2 2x8x10	70	56 main joists + 13 blocking boards + 1 spare
PT SYP No. 2 2x8x8	10	9 beam segments + 1 spare
PT SYP No. 2 2x8x12	17	15 beam segments + 2 spare
PT SYP No. 2 2x8x16	8	4 perimeter-rim halves + 3 long beam segments + 1 spare
PT 6x6x8 UC4A	4	18 short posts from 3 boards; fourth is defect/overage stock
PT 2x6x8	3	Eight 24-in corner braces; one spare board
PT 1x8x16 fascia	8	104 perimeter ft plus joints and overage
80-lb 4,000-psi concrete	128 bags	18 x 3.85 ft^3 / 0.60 = 116; rounded with 10% allowance
12-in x 48-in fiber forms	18	One 32-in shaft form per footing
Landscape fabric	800 ft^2	651 ft^2 covered area plus overlaps and 10%
3/4-in crushed rock	7.4 yd^3 / approx. 10.4 tons	3 in compacted plus waste/compaction allowance
Hardware and fasteners	See Sheet 5	Connector-specific purchasing schedule

QUANTITY NOTES

FOUNDATION	18 footings: 22-in round x 8-in pads, 12-in shafts, bottom at -38 in, top at +2 in.
GROUND	Cover 651 ft^2 with fabric and 3 in compacted angular crushed rock; maintain positive drainage.
TREATMENT	Use No. 2 or better PT SYP. Posts and members within 6 in of soil must be UC4A/ground contact.
STORAGE	Keep lumber stickered, shaded, and restrained before installation. Reject severe twist, split, or wane.

MATERIALS LIST - HARDWARE AND DECKING OPTIONS

Use one compatible corrosion-resistant connector and fastener system for treated lumber

Blue Springs, Missouri | July 16, 2026

HARDWARE AND FASTENERS

Hardware / fastener	Qty	Specified example / use
6x6 adjustable standoff bases	18	Simpson ABU66Z, ZMAX; 1-in standoff
5/8-in cast-in anchors	20	18 + 2 spare; HDG 5/8 x 10-in J/anchor bolts
Triple-2x beam / 6x6 caps	18	Simpson BCS2-3/6Z, ZMAX
2x8 joist hangers	170	Simpson LUS28Z; 168 required + 2 spare
BCS connector screws	200	Simpson SD10112; follow manufacturer schedule
ABU connector nails	250	0.148 x 1-1/2-in HDG; exact base schedule
ABU through-bolt sets	40	1/2-in HDG bolts, nuts, washers
Hanger header nails	500	0.148 x 3-in HDG connector nails
Hanger joist nails	750	0.148 x 1-1/2-in HDG connector nails
Built-up beam screws	225	Approved 3-in structural screws; staggered rows
Blocking / rim screws	800	#10 x 3-in ACQ-rated structural exterior screws
Knee-brace bolt sets	20	1/2 x 8-in HDG through-bolts, washers, nuts
Underside diagonal strap	1 x 100-ft roll	Simpson CS16-R or approved equivalent
Fascia screws	120	#9 x 2-1/2-in ACQ-rated fascia/trim screws
End-cut treatment	1 gal	Copper naphthenate meeting AWPA M4
Optional top tape	1 lot	2-in joist tape and 6-in beam tape

SELECT ONE DECKING PACKAGE

Both packages use the same board layout.
Option A is the recommended best value.

OPTION A - 5/4x6	Qty
5/4x6x16 PT SYP	100
5/4x6x12 PT SYP	4
5/4x6x10 PT SYP	4
#9 x 2-1/2-in deck screws	2,500

OPTION B - 2x6	Qty
2x6x16 PT SYP	100
2x6x12 PT SYP	4
2x6x10 PT SYP	4
#9/#10 x 3-in deck screws	2,500

COMPATIBILITY RULES

Use ZMAX/G185 HDG or stainless
Use listed connector fasteners only
Do not substitute ordinary screws
Follow current connector schedules
Predrill decking ends and knots